

Assignment Math review

 This is a preview of the published version of the quiz

Started: Apr 29 at 7:49pm

Quiz Instructions

It is very important that you master those basic skills.

There is a tutorial in the dropbox for most of the problems but TRY first

There are also 2 tutorials covering proportions and Sci Not in dropbox (pdfs also in dropbox)

For questions, I am available after our class.

2 attempts. the last one counts



Question 1 4.4 pts

Convert 1.2×10^{-3} to decimal notation.

☐

0.1200

☐

1.200

☐

0.0012

☐

0.00012

☐

0.0120



Question 2 4.4 pts

The shortest wavelength of visible light is approximately 400 nm. Express this wavelength in centimeters.

☐

400×10^{-11} cm

☐

4×10^{-7} cm

☐

4×10^{-11} cm



$4 \times 10^{-5} \text{ cm}$



$4 \times 10^{-9} \text{ cm}$



Question 3 4.4 pts

The following conversion equivalents are given:

$$1 \text{ gal} = 231 \text{ in}^3 \quad 1 \text{ ft} = 12 \text{ in} \quad 1 \text{ min} = 60 \text{ s}$$

A pipe delivers water at the rate of 95 gal/min. The rate in ft^3/s , is closest to:



0.19



0.21



0.15



0.14



0.17



Question 4 4.4 pts

0.0001776 can also be expressed as



$177.2 \times 10^7.$



$1.776 \times 10^{-4}.$



$1772 \times 10^5.$



$17.72 \times 10^4.$



$1.776 \times 10^{-3}.$



Question 5 4.4 pts

$0.00325 \times 10^{-8} \text{ cm}$ can also be expressed in mm as



$3.25 \times 10^{-10} \text{ mm.}$



$3.25 \times 10^{-12} \text{ mm.}$



$3.25 \times 10^{-11} \text{ mm.}$



$3.25 \times 10^{-8} \text{ mm.}$



$3.25 \times 10^{-9} \text{ mm.}$



Question 6 4.4 pts

Express $(4.3 \times 10^6)^{-1/2}$ in scientific notation.



4.8×10^{-4}



2.1×10^4



2.1×10^{-5}



2.1×10^3



Question 7 4.4 pts

A weight lifter can bench press 171kg. How many milligrams (mg) is this?



$1.71 \times 10^8 \text{ mg}$



$1.71 \times 10^9 \text{ mg}$



$1.71 \times 10^7 \text{ mg}$



$1.71 \times 10^6 \text{ mg}$



Question 8 4.4 pts

The wavelength of a certain laser is 0.35 micrometers (0.35 μm) Express this wavelength in nanometers.

(see equation sheet for prefix)



$3.5 \times 10^2 \text{ nm}$



$3.5 \times 10^4 \text{ nm}$



$3.5 \times 10^3 \text{ nm}$



$3.5 \times 10^1 \text{ nm}$



Question 9 4.4 pts

Convert 1.2×10^{-3} to decimal notation.



0.00012



1.200



0.0120



0.0012



0.1200



Question 10 4.4 pts

In addition to $1 \text{ m} = 39.37 \text{ in.}$, the following exact conversion equivalents are given below .

If a particle has a velocity of 8.4 miles per hour, its velocity, in m/s, is closest to

$1 \text{ mile} = 5280 \text{ ft}$

$1 \text{ ft} = 12 \text{ in}$



4.5 m/s.



3.8 m/s.



3.4 m/s.



3.0 m/s.



4.1 m/s.



Question 11 4.4 pts

A speed of 65 miles per hour is the same as



24 m/s



32 m/s



29 m/s



42 m/s



37 m/s



Question 12 4.4 pts

0.0001776 can also be expressed as

 17.72×10^4 . 177.2×10^7 . 1.776×10^{-4} . 1772×10^5 . 1.776×10^{-3} .

Question 13 4.4 pts

 0.00325×10^{-8} cm can also be expressed in mm as 3.25×10^{-10} mm. 3.25×10^{-12} mm. 3.25×10^{-11} mm. 3.25×10^{-8} mm.

 3.25×10^{-9} mm.

Question 14 4.4 pts

Convert a speed of 4.50 km/h to units of ft/min. (1.00 m = 3.28 ft)



886 ft/min



246 ft/min



165 ft/min



0.246 ft/min



82.3 ft/min



Question 15 4.4 pts

The following conversion equivalents are given:

$1 \text{ m} = 100$

cm

$1 \text{ in} = 2.54 \text{ cm}$

$1 \text{ ft} = 12 \text{ in}$

A bin has a volume of 1.5 m^3 . The volume of the bin, in ft^3 , is closest to:

35



53



47



41



59



Question 16 4.4 pts

One light-year is the distance light travels in one year. The speed of light is about 300,000 km/s (3×10^5 km/s). How far is 1 light-year? 9.46×10^{12} km

 3×10^5 km 1.08×10^9 km 1.8×10^7 km**Question 17 4.4 pts**

Light travels at a speed of 300,000 km/s. About how far is a light-year?



10 billion km



10 trillion km



10 million meters



300,000 km



10 million km

**Question 18 4.4 pts**

Suppose you know the speed of a spacecraft in kilometers per second. How would you calculate its speed in kilometers per hour?



Divide by 24.



Multiply by 24.



Multiply by 60 and then divide by 60 again.



multiply by 3600

**Question 19 4.4 pts**

How many seconds are in one year? (Calculate this, do not look it up.)

about 86 thousand (86,000, or 8.6×10^5)about 380 million (380,000,000, or 3.8×10^8)about 30 million (30,000,000, or 3×10^7)



about 3600 (3.6×10^3)



Question 20 4.4 pts

One light-hour is the distance light travels in one hour. The speed of light is about 300,000 km/s (3×10^5 km/s). If Jupiter is 0.72 light hours from the Sun, how far is this?



216 thousand km (2.16×10^5 km)



13 million km (1.3×10^7 km)



778 million km (7.78×10^8 km)



1.5 billion km (1.5×10^9 km)



Question 21 4.4 pts

The planet Mars is, on average, about 228 million km from the Sun. How long does it take light from the Sun to reach Mars? (Recall that the speed of light is about 300,000 km/s.)



about 1.52 light seconds



about 8.4 minutes



about 1.52 hours



about 12.7 minutes



Question 22 4.4 pts

Our solar system is located about 27,000 light-years from the galactic center. About how far does our solar system travel in one orbit?

hint: distance around a circle is $2\pi R$



85,000 light-years



170,000 light-years



54,000 light-years



100,000 light-years



Question 23 4.4 pts

The most basic of the sciences of physics, chemistry, and biology is (but of course!)

☐

biology.

☐

physics.

☐

chemistry.

☐

none of the above

Not saved

Submit Quiz